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Research Proposal

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Research Topic	Hybrid Machine Learning for Identifying Product and Service-Related Issues in E-Commerce Review Data		
Supervisor(s)	Mr. Y.M.S. Tharaka		

Agreement

I ,…M.R.M.Musharraf……, willing to submit the following Proposal format and I am confirming that this topic is a novel research and I am aware of the consequences occurs in the case of plagiarism found.

……. …....…09/10/2025……

Signature Date

Background

The exponential growth of e-commerce has generated massive volumes of user-generated content daily, with customer reviews serving as critical assets for both consumers and businesses. These reviews contain rich insights about product quality, service delivery, and customer satisfaction essential for competitive advantage. However, the sheer volume and unstructured nature of review data present significant challenges for extracting actionable insights [1]. Traditional manual analysis methods are inadequate for processing modern e-commerce feedback at scale, necessitating automated intelligent systems capable of identifying and categorizing product and service-related issues efficiently.

Sentiment analysis has evolved from simple polarity classification to sophisticated multi-dimensional techniques. Early approaches focused on document-level classification, categorizing reviews as positive, negative, or neutral [2]. These methods failed to capture aspect-specific opinions common in e-commerce where customers simultaneously express different sentiments about various product features [3]. This limitation drove the development of Aspect-Based Sentiment Analysis (ABSA), enabling fine-grained opinion mining at the feature level [4]. Recent ABSA implementations using RoBERTa-based models have achieved 90% accuracy in identifying purchase process aspects and their sentiments [3].

Traditional machine learning techniques including SVM, Naive Bayes, and Random Forests have been extensively applied using handcrafted features such as TF-IDF and n-grams [1], [39]. While computationally efficient and interpretable, these methods struggle with semantic relationships and contextual information [2], [33]. Deep learning revolutionized sentiment analysis with LSTM and BiLSTM networks capturing sequential dependencies [12], and CNNs extracting local textual patterns [11]. Attention mechanisms further enhanced performance, with Hierarchical Attention Networks achieving 91.70% accuracy on aspect extraction [13].

Transformer-based architectures, particularly BERT, marked a paradigm shift through bidirectional attention mechanisms that capture context from both directions [41], [42]. Fine-tuned BERT models have achieved accuracies up to 97.3% on review datasets [41]. Variants including RoBERTa and ALBERT enhanced performance through improved training procedures [3], [7]. Domain-specific BERT adaptations combined with CNN-BiLSTM achieved F1-scores of 0.941, significantly outperforming traditional approaches [42].

Hybrid approaches combining multiple techniques have emerged to leverage complementary strengths. Hashmi and Yayilgan [5] achieved 89% accuracy using hybrid BERT with explainable AI components through LIME and attention mechanisms. Kumar et al. [6] demonstrated F1-scores of 0.92 through ensemble-deep learning combining CNN-LSTM with large language models. Hybrid feature engineering combining multiple extraction methods has significantly improved aspect-level classification [1], with advanced term weighting and optimization achieving 25-32% improvements over baselines [9].

Despite advances, several challenges persist. Fake review detection requires sophisticated ABSA techniques considering product types and aspect inconsistencies [7]. Customer dissatisfaction detection faces challenges without ground truth labels, necessitating semi-supervised approaches [6]. Multimodal integration combining text with images requires careful architectural design [10]. Model interpretability remains critical for business adoption, requiring explainable AI techniques [5].

Critical gaps motivate this research. First, most approaches focus on sentiment polarity rather than specifically identifying and categorizing product and service issues [2], [8]. Second, systematic investigation of optimal combinations of traditional ML, deep learning, and transformer-based approaches remains limited [9], [14]. Third, comprehensive end-to-end frameworks integrating multiple components are lacking [13]. Fourth, models often fail to generalize across diverse e-commerce domains [1], [6]. Finally, high-performing deep learning models lack interpretability while transparent traditional methods sacrifice accuracy [5], [20].

These gaps necessitate a comprehensive hybrid machine learning approach specifically designed for identifying product and service-related issues in e-commerce reviews. By combining complementary strengths of traditional ML, deep learning, and transformer-based models, this research aims to achieve superior performance while maintaining interpretability and generalizability across diverse e-commerce contexts, advancing the state-of-the-art in automated issue identification through systematic hybrid methodology.

Literature Review

Traditional Machine Learning for Sentiment Analysis

Traditional machine learning approaches employ feature engineering techniques like Bag-of-Words, TF-IDF, and n-grams followed by classification algorithms [32], [39]. SVM has been widely adopted, achieving 83-94% accuracy on various datasets [33], [37], while Naive Bayes and Random Forests demonstrate efficiency and robustness [32], [34]. However, these methods struggle to capture semantic relationships, fail to account for word order and context, and require extensive manual feature engineering [2], motivating the shift toward deep learning.

Deep Learning Approaches for Sentiment Analysis

Deep learning enables automatic feature learning and captures complex linguistic patterns. LSTM networks address vanishing gradients and capture long-range dependencies [12]. Gondhi [12] applied optimized LSTM with hybrid mutation algorithms (HMEWA) to e-commerce clothing reviews, achieving high accuracy. Kuppusamy [11] demonstrated that hybrid CNN-BiLSTM architectures achieved 93.6% accuracy by combining local feature extraction with sequential processing. Attention mechanisms enable focus on relevant text portions [8], [13]. Kalaivaani [13] implemented HAN-CNN achieving 91.70% accuracy with 91.20% F1-score. Unsupervised attention frameworks extract OACIS quintuples through clustering and negative sampling [8], achieving supervised-level performance with reduced annotation.

Transformer-Based Models and Transfer Learning

BERT's bidirectional attention enables nuanced linguistic understanding [41]. Areshey and Mathkour [41] demonstrated fine-tuned BERT achieved 97.3% accuracy, showing 46% improvement over traditional approaches. RoBERTa improved training procedures [3]. Davoodi et al. [3] achieved 90% accuracy using RoBERTa for ABSA on 3,500 Trustpilot reviews, identifying purchase process aspects. Domain-specific BERT adaptations show superior performance [42]. Prottasha et al. [42] demonstrated domain-specific Bangla-BERT with CNN-BiLSTM achieved F1-scores of 0.941, outperforming Word2Vec, GloVe, FastText, and multilingual BERT.

Aspect-Based Sentiment Analysis

ABSA enables fine-grained opinion mining by identifying specific features and sentiments. Yadav [4] developed an ABSA framework using WordNet for aspect extraction on Amazon reviews for personalized recommendations. Yiran and Srivastava [18] applied LDA with ML classifiers to 1,000 Amazon iPhone X reviews for unsupervised aspect discovery. Neural approaches jointly learn extraction and classification [8]. Sharma and Saraswat [20] demonstrated that integrating domain ontologies with BiLSTM achieved 91.57% accuracy. Joint models avoid pipeline error propagation, achieving supervised performance with less annotation [8].

Hybrid Machine Learning Approaches

Hybrid approaches leverage complementary strengths of multiple techniques. The Amazon headphone study [1] demonstrated combining traditional features with neural embeddings improved aspect-level classification using SVM, KNN, Random Forests, Naive Bayes, Decision Trees, and Logistic Regression. Advanced term weighting using LF-MICF with IGWO achieved 25-32% improvements [9]. Kumar et al. [6] proposed hybrid ensemble-deep learning combining CNN-LSTM with LLMs through pseudo-labeling, achieving F1-scores of 0.92 for dissatisfaction detection without ground truth. Kuppusamy [11] showed CNN-BiLSTM hybrids achieved 93.6% accuracy. HAN-CNN combinations enable hierarchical attention [13], achieving 91.70% accuracy. Hashmi and Yayilgan [5] developed hybrid BERT with LIME, achieving 89% accuracy while maintaining interpretability. Fused networks combining Chinese BERT with CNN and BiLSTM showed multilingual effectiveness [16]. Two-channel models using MWVH aggregate Word2Vec, GloVe, and FastText properties [17].

Specialized Applications in E-Commerce

ABSA-based approaches detect fake reviews by analyzing aspect-sentiment coherence [7]. Kumar et al. [6] used semi-supervised learning for dissatisfaction detection, identifying product defects, delivery problems, service issues, and pricing concerns with F1-scores of 0.92. Yadav [4] developed aspect-level sentiment frameworks for personalized recommendations. Deep learning-based systems improve satisfaction [15]. Quality management benefits from automated issue identification with explainable AI ensuring interpretability [5], [6].

Multimodal Approaches and Innovations

Li et al. [10] introduced a text-to-image integration approach for sentiment classification using 5,000 Amazon reviews. Their CNN-SVM model, enhanced with Word2Vec-TextRank and generative AI, achieved 97.13% accuracy. Visual representations improved classification by adding semantic and contextual cues. Key challenges include feature alignment, weighting strategies, and computational complexity. Fusion techniques such as early, late, and hybrid fusion with attention mechanisms were also explored.

Explainability and Interpretability

Attention mechanisms provide interpretability by highlighting relevant regions [5], [13]. LIME generates local explanations by approximating complex models [5]. Hashmi and Yayilgan [5] integrated LIME with BERT, achieving 89% accuracy with interpretability. Simple models offer transparency but sacrifice accuracy, while complex models achieve performance but lack interpretability [5], [20]. Domain ontologies enhance both performance and interpretability [20]. Sharma and Saraswat [20] showed ontologies with BiLSTM achieved 91.57% accuracy.

Research Gaps and Future Directions

Critical gaps motivate this research. First, specific focus on identifying and categorizing product and service-related issues remains limited [2], [8]. Most approaches classify sentiment polarity rather than extracting structured issue information. Second, systematic investigation of optimal combinations of traditional ML, deep learning, and transformers is insufficient [9], [14]. Questions remain about optimal architectures and fusion strategies. Third, models trained on specific categories fail to generalize across domains [1], [6]. Domain shift requires significant labeled data for adaptation. Fourth, transformer computational requirements limit resource-constrained deployment [11], [13]. Memory and latency present barriers for real-time applications. Fifth, most research focuses on English with limited multilingual work [3], [16]. These gaps highlight the need for comprehensive hybrid frameworks for identifying product and service-related issues in e-commerce reviews, addressing generalization, efficiency, and interpretability.

Technical Novelty & Justification of Hybrid Model

The proposed hybrid model combines:

- Classical ML (SVM, Random Forest) for robust feature interpretation,
- Deep learning (Bi-LSTM + BERT) for rich contextual embeddings,
- Unsupervised learning (LDA) for automatic aspect discovery.

This three-tier integration is novel because no existing study simultaneously uses LDA-guided aspect clustering with BERT-based contextual representation and classical ML classifiers for multi-class product/service issue detection. This reduces dependency on large labeled datasets, improves interpretability, and enhances performance on noisy, imbalanced real-world review data.

Gaps in Existing Research – Detailed Study

Even though research on machine learning (ML) and natural language processing (NLP) for e-commerce reviews has made good progress, there are still important gaps. These gaps make it hard to fully identify and sort product and service issues from reviews. This section explains these gaps in detail, using examples from key studies, and shows why a hybrid ML model is needed to fix them.

One big gap is that most studies focus on basic sentiment checks—like if a review is good, bad, or neutral—instead of spotting specific issues. For example, a 2022 study from Taylor & Francis used Chinese BERT with deep networks to analyze 100,000 e-commerce reviews and got high accuracy, but it only did simple yes/no classification. This misses details like product problems (e.g., low battery life) or service complaints (e.g., slow shipping). Another 2023 Taylor & Francis paper on deep learning for product suggestions looked at general feelings to recommend items, but it did not break down issues into groups like quality or price. In e-commerce, where returns can reach 30%, businesses need tools to find exact problems, not just overall moods. Without this, companies miss chances to improve products and keep customers happy.

Another gap is the lack of models that mix guided (supervised) and unguided (unsupervised) methods to handle real data. Many studies need labeled reviews, which take time and money to prepare. A 2024 study from Emerald Insight by Sharma and Saraswat combined deep learning with word lists and achieved 91.57% accuracy for aspect-based sentiment analysis (ABSA), but it relied on ready-made labels and did not use unguided tools like grouping topics for new data. A 2023 Wiley paper by Kuppusamy and Selvaraj proposed a mixed deep learning model for ABSA with networks like CNN and RNN, but it did not handle messy or incomplete reviews common on sites like Amazon. Without unguided parts, these models cannot scale to the millions of unlabeled reviews added daily, leaving out useful insights from data.

Handling reviews in different languages or with noise is also not well covered. E-commerce reviews often mix languages, especially in places like Sri Lanka where online shopping grows 15% each year. A 2020 Emerald Insight study used a mix of rules and word lists for app reviews, which worked for English but did not test other languages. A 2024 Elsevier paper by Li et al. turned review words into images with CNN-SVM and got 97.13% accuracy on Amazon data, but it skipped noise from short or casual reviews. This gap matters in global markets, where ignoring non-English or messy text leads to incomplete views of customer issues.

Models also lack good scaling and easy-to-understand results. Deep learning tools, like the two-part model in a 2024 Elsevier paper by Wang et al. (95.14% accuracy on international reviews), need strong computers and big data sets, which small businesses cannot afford. They often do not explain why they flagged an issue, making it hard to trust. A 2025 Springer paper by Prajapati et al. used RoBERTa for 92.53% accuracy on Trustpilot reviews to spot shipping problems, but it did not show clear reasons or work well on huge data without extra help. Mixed models like one in a 2023 Springer study (93-94% accuracy) added simple explanations, but they still focused on feelings rather than many issue types.

Last, few studies test models in real-time or across different platforms. A 2023 ACM paper on mixed ML for suggestions worked on certain data but did not try it on multiple sites like Amazon and Flipkart. 2025 Springer papers by Tu et al. and Zaninotto et al. used topic grouping and analysis for trends and complaints, finding things like delivery issues, but they did not mix this with ML to predict problems ahead.

These gaps mean current work is okay for basic checks but not enough for detailed, easy-to-use issue spotting. The planned hybrid model—using simple ML like SVM or Random Forest with deep tools like Bi-LSTM or BERT, plus grouping like LDA—will fix this. It will handle unlabeled and mixed-language data, sort issues into groups (e.g., product faults vs. service delays), and give clear explanations. This will help e-commerce in places with limited tools, like Sri Lanka, to cut complaints and grow better.

Problem Statement and Research Questions

Problem Statement

E-commerce platforms generate millions of unstructured customer reviews containing critical product and service-related issues. Existing approaches either lack contextual understanding (traditional ML), require expensive labeled data (deep learning), or fail to perform multi-class issue classification effectively. This research proposes a novel hybrid machine learning model integrating classical ML, deep learning, and unsupervised techniques to accurately detect and classify product and service-related issues in e-commerce reviews.

Research Questions

RQ01: How can a hybrid model combining classical ML, deep learning, and unsupervised methods improve multi-class issue detection compared to single-approach methods?

RQ02: Which NLP preprocessing techniques are most effective for optimizing the hybrid model's performance on e-commerce review data?

RQ03: How does the hybrid model perform across different issue categories, and what optimizations enhance its accuracy?

RQ04: Can the hybrid model handle real-world challenges like noisy text, imbalanced data, and diverse platform reviews?

RQ05: What actionable insights can the model provide to help businesses reduce specific issues and improve customer satisfaction?

Research Objectives and Goals

General Objective

To develop and evaluate a hybrid machine learning model that accurately identifies and classifies product-related and service-related issues from e-commerce review data, providing actionable insights for improving customer satisfaction.

Specific Objectives

Objective 1: Develop a hybrid machine learning model combining classical ML (SVM, Random Forest), deep learning (Bi-LSTM, BERT), and unsupervised methods (LDA) for multi-class issue detection in e-commerce reviews. *(Addresses RQ01)*

Objective 2: Identify and implement optimal NLP preprocessing techniques (tokenization, lemmatization, noise removal) to maximize hybrid model performance. *(Addresses RQ02)*

Objective 3: Evaluate model performance across different issue categories using accuracy, precision, recall, and F1-score, and implement optimization strategies. *(Addresses RQ03)*

Objective 4: Assess model robustness in handling real-world challenges including noisy text, imbalanced data, and diverse platform reviews. *(Addresses RQ04)*

Objective 5: Generate actionable insights and recommendations to help e-commerce businesses systematically address specific issues and improve customer satisfaction. *(Addresses RQ05)*

Proposed Research Method

The proposed research method aims to develop and test a hybrid machine learning model to identify and classify product-related and service-related issues from e-commerce review data. The following steps outline the systematic approach:

1. **Data Collection:** Gather large-scale e-commerce review datasets from publicly available sources, such as Amazon Product Reviews or Flipkart datasets from Kaggle, targeting approximately 10000 reviews. These datasets will include diverse product categories and multilingual reviews to reflect real-world variability.
    - Public datasets: Amazon 2018, Flipkart, Multi-domain (Kaggle)
    - Ethical web scraping: Daraz.lk, Kapruka, Takas.lk (2024–2025 reviews, Sinhala–English mixed)
      Tools: BeautifulSoup, Scrapy, Selenium (robots.txt compliant)

2. **Data Preprocessing:** Prepare the text data using natural language processing (NLP) techniques. This includes tokenization (breaking text into words), stop-word removal (eliminating common words like "the" or "and"), lemmatization (reducing words to base forms), and noise reduction (removing typos or special characters). The goal is to clean the data for effective analysis, addressing Research Question 2.

3. **Data Labeling:** Label the preprocessed reviews into predefined issue categories (e.g., product quality, delivery, customer service, pricing) using a semi-automated approach. Initial labeling will use rule-based methods and keyword matching, followed by manual validation by the researcher to ensure accuracy, preparing data for supervised learning components.

4. **Model Development:** Design a hybrid machine learning model combining classical techniques (e.g., Support Vector Machines or Random Forest) for feature extraction with deep learning architectures (e.g., Bi-LSTM or BERT) for contextual understanding. Incorporate unsupervised Latent Dirichlet Allocation (LDA) for clustering unlabeled data, enhancing the model’s ability to handle diverse reviews, as outlined in Research Objective 2.

5. **Model Training and Optimization:** Split the dataset into 80% training and 20% testing sets. Train the hybrid model using Python with libraries like TensorFlow and Scikit-learn, optimizing hyperparameters (e.g., learning rate, batch size) to improve performance across issue categories, addressing Research Question 3.

6. **Model Evaluation:** Assess the model's performance using metrics such as accuracy, precision, recall, and F1-score. Conduct cross-validation to ensure reliability and test on noisy or multilingual subsets to evaluate real-world applicability, responding to Research Question 4.

7. **Insight Generation and Validation:** Analyze model outputs to extract actionable insights (e.g., common quality issues, frequent delivery delays) and validate findings with a small sample of e-commerce stakeholders (e.g., local sellers) to ensure practicality, aligning with Research Question 5.

8. **Tools and Environment:** Use Python for implementation, with NLTK for NLP tasks, TensorFlow/Keras for deep learning, and Scikit-learn for classical ML. Cloud computing (e.g., Google Colab) will support training on large datasets.

Expected Outcome

This research aims to develop a hybrid machine learning model for identifying and classifying product and service-related issues from e-commerce review data. The expected outcomes are:

Outcome 1: Improved Multi-Class Issue Detection Accuracy - The hybrid model integrating classical ML (SVM, Random Forest), deep learning (Bi-LSTM, BERT), and unsupervised methods (LDA) is expected to achieve 90-95% accuracy in detecting and categorizing specific product and service issues, outperforming existing single-approach methods. *(Addresses RQ01)*

Outcome 2: Optimized NLP Preprocessing Framework - A refined set of effective NLP techniques (tokenization, lemmatization, noise removal, feature extraction) that enables robust preprocessing of e-commerce review data for maximum model performance. *(Addresses RQ02)*

Outcome 3: Consistent Cross-Category Performance -Demonstrated consistent performance across diverse issue categories (product quality, defects, packaging, delivery delays, customer support) with validated optimization strategies based on accuracy, precision, recall, and F1-score metrics. *(Addresses RQ03)*

Outcome 4: Real-World Robustness and Scalability - A robust model capable of handling noisy text, informal language, imbalanced datasets, and reviews from diverse e-commerce platforms, proving practical applicability and scalability. *(Addresses RQ04)*

Outcome 5: Actionable Business Intelligence - Specific, data-driven recommendations identifying frequent product defects and service bottlenecks to help e-commerce businesses reduce issues, lower return rates, and enhance customer satisfaction. *(Addresses RQ05)*

Research Plan and Time Table

Activity	1st Month				2nd Month				3rd Month				4th Month				5th Month				6th Month			
	1st Week	2nd Week	3rd Week	4th Week	1st Week	2nd Week	3rd Week	4th Week	1st Week	2nd Week	3rd Week	4th Week	1st Week	2nd Week	3rd Week	4th Week	1st Week	2nd Week	3rd Week	4th Week	1st Week	2nd Week	3rd Week	4th Week
Literature Review (Gap analysis, objective setting)																								
Data Collection (Dataset gathering, preprocessing)																								
Model Development (Hybrid model design, coding)																								
Model Training (Training, optimization)																								
Evaluation (Testing, stakeholder validation)																								
Documentation (Report writing, dissemination)																								

Table 1: Research Plan and Time Table

Proposed Budget

The primary costs for this research will be internet access, printing, and dissemination expenses.

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Any Other Relevant Information